PROJECT COLLECTION

REZA BABAPIRI

Embedded Systems & Robotics

Microcontrollers, sensing and mobile robots

1 selected documented project

Project context / Methods / Evidence / Limitations

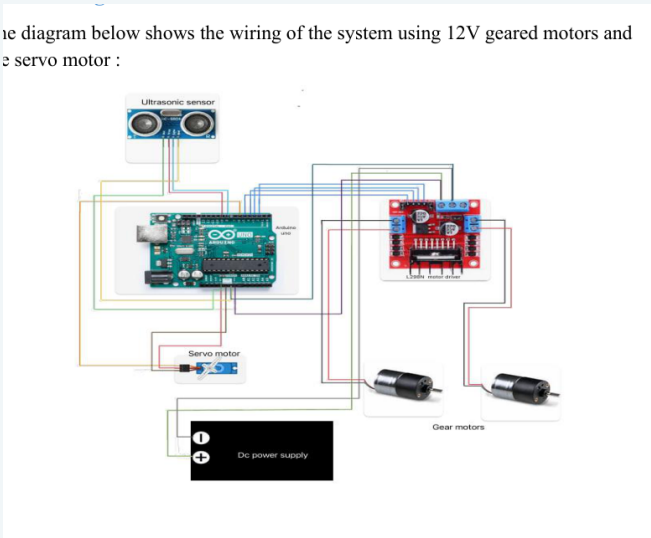
reza-babapiri.github.io

Academic work and anonymised professional experience | Source-derived summaries

EMBEDDED HARDWARE · MOBILE ROBOTICS

Autonomous Obstacle-Avoiding Robot

Arduino Uno, ultrasonic sensing, servo scanning and DC motor driving



ENGINEERING SUMMARY

Documented an obstacle-avoiding mobile robot assembled for an embedded systems course by a two-person team. A servo-mounted ultrasonic sensor scans the front and sides and an Arduino Uno drives motors through an L298N module.

METHOD / CONTRIBUTIONS

- Integrated Arduino Uno, HC-SR04 ultrasonic sensor, servo motor and dual DC geared motors.
- Implemented repeated distance sampling with out-of-range filtering and averaging.
- Used motor-control functions and threshold-based obstacle detection to stop, reverse and turn.

RESULTS & EVIDENCE

- Report documents a 40 cm front obstacle threshold and timing-based drive commands.
- Full Arduino source and a wiring diagram are present in the supplied group report.

SCOPE / LIMITATION

The report describes choosing the clearer side, but the supplied final loop alternates turning with turnCounter. This inconsistency is disclosed rather than presenting unverified path planning.

SUPPORTING RECORD

15-page group report; wiring diagram, Arduino C++ source, component list.